

# Alexey Ananyevskiy

Curriculum Vitae, September 2026

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🌐 [www.aananyevskiy.github.io](http://www.aananyevskiy.github.io)

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## Employment

- 2022– **Heisenberg Position**, Ludwig-Maximilians-Universität München, Munich, Germany
- 2017–2022 **Deputy Director**, EIMI, St. Petersburg, Russia
- 2017–2022 **Senior Researcher**, PDMI RAS, St. Petersburg, Russia
- 2017–2019 **Senior Researcher**, Chebyshev Laboratory at SPbU, St. Petersburg, Russia
- 2015–2019 **Deputy Head**, Chebyshev Laboratory at SPbU, St. Petersburg, Russia
- 2013–2017 **Postdoc**, Chebyshev Laboratory at SPbU, St. Petersburg, Russia
- 2010–2013 **Lecturer**, Algebra Department at SPbU, St. Petersburg, Russia

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## Education

- 2010–2013 **Ph.D. in Mathematics**, *SPbU, St. Petersburg, Russia*  
Thesis: "On the algebraic K-theory of some varieties and related problems"  
Doctoral advisor: Ivan Panin
- 2005–2010 **Specialist in Mathematics**, *SPbU, St. Petersburg, Russia*  
Thesis: "Generalization of Colliot-Thélène and Sansuc theorem on principal T-bundles"

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## Research visits

- Oct – Nov 2026 SLMATH, Berkeley, USA (planned)
- May – Jun 2022 Isaac Newton Institute, Cambridge, UK (online due to travel restrictions)
- Mar 2020 Isaac Newton Institute, Cambridge, UK; suspended due to the COVID-19 pandemic
- Oct 2018 University of Oslo, Oslo, Norway
- Feb – Mar 2018 University of Oslo, Oslo, Norway
- Jan – Apr 2017 Mittag-Leffler Institute, Stockholm, Sweden
- Nov 2016 Universität Osnabrück, Osnabrück, Germany
- Jan – Apr 2015 Institute for Advanced Study, Princeton, USA
- Jun 2014 Universität Duisburg-Essen, Essen, Germany
- May 2013 Fields Institute, Toronto, Canada
- Jul – Aug 2011 Universität Bielefeld, Bielefeld, Germany
- Jun – Jul 2009 Universität Bielefeld, Bielefeld, Germany

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## Areas of specialization

Motivic Homotopy Theory, Vector bundles, Algebraic K-theory, Homogeneous Varieties, Algebraic Geometry

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## Publications

- 23. Alexey Ananyevskiy, Alexander Samokhin, *Semiorthogonal decompositions for families of twisted flag varieties*, <https://arxiv.org/abs/2609.05203>

22. Alexey Ananyevskiy, Nikita Geldhauser, *Chow rings of quasi-split geometrically almost simple algebraic groups*, <https://arxiv.org/abs/2408.09390>
21. Alexey Ananyevskiy, Marc Levine, *Combing a hedgehog over a field*, *Algebra and Number Theory* 19:12 (2025), 2409–2431, <https://doi.org/10.2140/ant.2025.19.2409>
20. Alexey Ananyevskiy, Elden Elmanto, Oliver Röndigs, Maria Yakerson, *On the motivic Adams conjecture*, *Journal of Topology* 18:2 (2025), e70026, <https://doi.org/10.1112/topo.70026>
19. Alexey Ananyevskiy, *On the  $\mathbb{A}^1$ -Euler characteristic of the variety of maximal tori in a reductive group*, *IMRN*, 2022:19 (2022), 15390–15409, <https://doi.org/10.1093/imrn/rnab156>
18. Alexey Ananyevskiy, *Thom isomorphisms in triangulated motivic categories*, *Algebraic and Geometric Topology*, 21:4 (2021), 2085–2106, <https://doi.org/10.2140/agt.2021.21.2085>
17. Alexey Ananyevskiy, Grigory Garkusha, Ivan Panin, *Cancellation theorem for framed motives of algebraic varieties*, *Advances in Mathematics*, 383 (2021), 107681, <http://doi.org/10.1016/j.aim.2021.107681>
16. Alexey Ananyevskiy, *SL-oriented cohomology theories*, in *Motivic Homotopy Theory and Refined Enumerative Geometry*, *Contemporary Mathematics*, 745 (2020), 1–20, <https://doi.org/10.1090/conm/745>
15. Alexey Ananyevskiy, Oliver Röndigs, Paul Arne Østvær, *On very effective hermitian K-theory*, *Mathematische Zeitschrift*, 294 (2020), 1021–1034, <https://doi.org/10.1007/s00209-019-02302-z>
14. Alexey Ananyevskiy, Alexander Neshitov, *Framed and MW-transfers for homotopy modules*, *Selecta Mathematica*, 25:26 (2019), <https://doi.org/10.1007/s00029-019-0472-0>
13. Alexey Ananyevskiy, Andrei Druzhinin, *Rigidity for linear framed presheaves and generalized motivic cohomology theories*, *Advances in Mathematics*, 333 (2018), 423–462, <https://doi.org/10.1016/j.aim.2018.05.013>
12. Alexey Ananyevskiy, *On the zeroth stable  $\mathbb{A}^1$ -homotopy group of a smooth curve*, *Journal of Pure and Applied Algebra*, 222:10 (2018), 3195–3218, <https://doi.org/10.1016/j.jpaa.2017.12.001>
11. Alexey Ananyevskiy, Marc Levine, Ivan Panin, *Witt sheaves and the  $\eta$ -inverted sphere spectrum*, *Journal of Topology*, 10:2 (2017), 370–385, <https://doi.org/10.1112/topo.12015>
10. Alexey Ananyevskiy, *Stable operations and cooperations in derived Witt theory with rational coefficients*, *Annals of K-theory*, 2:4 (2017), 517–560, <https://doi.org/10.2140/akt.2017.2.517>
9. Alexey Ananyevskiy, *On the zeroth stable  $\mathbb{A}^1$ -homotopy group of a smooth projective variety*, *Journal of Mathematical Sciences*, 222:4 (2017), 367–369 (translated from Russian), <https://doi.org/10.1007/s10958-017-3306-7>
8. Alexey Ananyevskiy, *On the push-forwards for motivic cohomology theories with invertible stable Hopf element*, *Manuscripta Mathematica*, 150:1–2 (2016), 21–44, <https://doi.org/10.1007/s00229-015-0799-6>
7. Alexey Ananyevskiy, *On the relation of special linear algebraic cobordism to Witt groups*, *Homology, Homotopy and Applications*, 18:1 (2016), 205–230, <https://doi.org/10.4310/HHA.2016.v18.n1.a11>
6. Alexey Ananyevskiy, *The special linear version of the projective bundle theorem*, *Compositio Mathematica*, 151:3 (2015), 461–501, <https://doi.org/10.1112/S0010437X14007702>

5. Alexey Ananyevskiy, Asher Auel, Skip Garibaldi, Kirill Zainoulline, *Exceptional collections of line bundles on projective homogeneous varieties*, Advances in Mathematics, 236 (2013), 111-130, <https://doi.org/10.1016/j.aim.2012.12.016>
4. Alexey Ananyevskiy, *Relationship between algebraic MSL-cobordisms and derived Witt groups*, Doklady Mathematics, 87:1 (2013), 76-78 (translated from Russian), <http://doi.org/10.1134/S1064562413010286>
3. Alexey Ananyevskiy, *On the algebraic K-theory of some homogeneous varieties*, Documenta Mathematica, 17 (2012), 167-193, <https://www.emis.de/journals/DMJDMV/vol-17/07.html>
2. Alexey Ananyevskiy, Nikolai Vavilov, Sergey Sinchuk, *Overgroups of  $E(m; R) \otimes E(n; R)$ . I. Levels and normalisers*, St. Petersburg Mathematical Journal, 23:5 (2012), 819-849 (translated from Russian), <http://doi.org/10.1090/S1061-0022-2012-01219-7>
1. Alexey Ananyevskiy, Nikolai Vavilov, Sergey Sinchuk, *On the description of overgroups of  $E(m; R) \otimes E(n; R)$* , Journal of Mathematical Sciences, 161:4 (2009), 461-473 (translated from Russian), <https://doi.org/10.1007/s10958-009-9576-y>

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## Selected talks

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| September 10, 2026 | <i>Combing hedgehogs over a field</i> , conference <i>Geometria in Bicocca 2026</i> , University of Milano-Bicocca, Milan, Italy                                                                           |
| September 08, 2026 | <i>Combing hedgehogs over a field</i> , DMV Annual Meeting, Konstanz University, Konstanz, Germany                                                                                                         |
| December 05, 2025  | <i>On semiorthogonal decompositions for twisted flag varieties</i> , <i>Complex algebraic geometry seminar</i> , Università degli Studi di Milano, Milan, Italy                                            |
| September 04, 2025 | <i>On the semiorthogonal decompositions for twisted flag varieties</i> , Workshop <i>Semiorthogonal decompositions for representations of algebraic groups</i> , Universität Bielefeld, Bielefeld, Germany |
| March 17, 2025     | <i>Nowhere vanishing sections of vector bundles revisited</i> , Conference <i>Motivic homotopy theory</i> , Universität Regensburg, Regensburg, Germany                                                    |
| October 22, 2024   | <i>Combing a hedgehog over a field</i> , <i>Algebraic geometry seminar</i> , University of Zurich, Zurich, Switzerland                                                                                     |
| January 15, 2024   | <i>Combing a hedgehog over a field</i> , <i>Algebra seminar</i> , Charles University, Prague, Czech Republic                                                                                               |
| November 23, 2023  | <i>Combing a hedgehog over a field</i> , <i>Kolloquium Geometrie und Arithmetik</i> , Johannes Gutenberg Universität Mainz, Mainz, Germany                                                                 |
| October 19, 2023   | <i>On the motivic Adams conjecture</i> , <i>AG seminar</i> , Universität Regensburg, Regensburg, Germany                                                                                                   |
| September 28, 2023 | <i>Non-Vanishing Sections of Vector Bundles and Chern Classes</i> , <i>ESAGA seminar</i> , Universität Duisburg-Essen, Essen, Germany                                                                      |
| June 22, 2023      | <i>On the existence of non-vanishing sections of vector bundles</i> , <i>Séminaire d'arithmétique à Lyon</i> , ENS de Lyon, Lyon, France                                                                   |
| May 10, 2023       | <i>Non-Vanishing Sections of Algebraic Vector Bundles and Trivial Chern Classes</i> , <i>Colloquium of Institut für Mathematik</i> , Universität Osnabrück, Osnabrück, Germany                             |
| November 17, 2022  | <i>On the existence of a nowhere vanishing section for a vector bundle with trivial top Chern class</i> , <i>Oberseminar zur Algebra und Zahlentheorie</i> , Universität Augsburg, Augsburg, Germany       |

- October 27, 2022 *On the existence of a nowhere vanishing section for a vector bundle with trivial top Chern class*, seminar *Motivic algebraic topology*, LMU, Munich, Germany
- July 25 – 27, 2021 *Motivic homotopy theory and vector bundles*, summer school *Algebra and Geometry*, Yaroslavl, Russia
- May 20, 2021 *An introduction to motivic homotopy theory*, Mini-workshop on Algebraic Geometry, EIMI, St. Petersburg, Russia
- April 29, 2021 *On the  $\mathbb{A}^1$ -Euler characteristic of the variety of maximal tori*, Motivic Algebraic Topology Seminar, LMU, online
- January 12, 2021 *On the  $\mathbb{A}^1$ -Euler characteristic of the variety of maximal tori*, Shafarevich Seminar, online
- September 9, 2020 *On the  $\mathbb{A}^1$ -Euler characteristic of the variety of maximal tori*, Conference *Motivic Geometry*, CAS, Oslo, Norway (moved to online)
- June 20, 2019 *Motivic cellular structure on some homogeneous varieties*, Conference *Number Theory and Algebraic Geometry (In memoriam of Alexei Zykin)*, Independent University of Moscow, Moscow, Russia
- May 7, 2019 *SL-oriented cohomology theories*, Conference *Algebro-geometric and homotopical methods*, Institute Mittag-Leffler, Stockholm, Sweden
- February 19, 2019 *Motives of smooth algebraic varieties*, Shafarevich Seminar, MI RAS, Moscow, Russia
- October 24, 2018 *Non-vanishing sections of algebraic vector bundles and characteristic classes*, Geometry and Topology Seminar, University of Oslo, Oslo, Norway
- June 29, 2018 *On the linear categories of motives*, Workshop *Motivic homotopy groups of spheres III*, Technische Universität Berlin, Berlin, Germany
- May 14, 2018 *SL-oriented cohomology theories*, Workshop *Motivic homotopy theory and refined enumerative geometry*, Universität Duisburg-Essen, Essen, Germany
- March 6, 2018 *Framed and Milnor-Witt transfers for homotopy modules*, Geometry and Topology Seminar, University of Oslo, Oslo, Norway
- February 7, 2017 *Rigidity for framed presheaves and motivic homotopy groups*, Mittag-Leffler Seminar, Institute Mittag-Leffler, Stockholm, Sweden
- November 29, 2016 *Algebraic K-theory of some homogeneous varieties*, Kollegseminar, Universität Osnabrück, Osnabrück, Germany
- November 22, 2016 *Rigidity for motivic stable homotopy sheaves*, Topology Seminar, Universität Osnabrück, Osnabrück, Germany
- June 24, 2016 *On the zeroth stable  $\mathbb{A}^1$ -homotopy group of a smooth curve*, Workshop on motivic homotopy groups of spheres II, Universität Duisburg-Essen, Essen, Germany
- February 3, 2016 *Stable operations in derived Witt theory and motivic Serre finiteness*, Workshop *Algebraic Cobordism and Projective Homogeneous Varieties*, Oberwolfach, Germany
- September 25, 2015 *Operations in derived Witt theory*, Minisymposium *Motivic Homotopy Theory and its application to problems in Algebra and Algebraic Geometry*, Hamburg, Germany
- June 20, 2014 *Special linear oriented cohomology theories*, Workshop *Oriented theories and symplectic cobordism*, Universität Duisburg-Essen, Essen, Germany

- September 28, 2013 *Cohomology theories with invertible Hopf element*, Workshop *Quantum and motivic cohomology, Fano varieties and mirror symmetry*, Euler Institute, St. Petersburg, Russia
- May 8, 2013 *SL-oriented cohomology theories*, Spring School and Workshop on Torsors, Motives and Cohomological Invariants, The Fields Institute for Research in Mathematical Sciences, Toronto, Canada
- April 23, 2013 *SL-oriented cohomology theories*, Shafarevich Seminar, MI RAS, Moscow, Russia
- September 20, 2012 *The special linear version of the projective bundle theorem*, Conference *Algebraic groups and related structures*, Euler Institute, St. Petersburg, Russia

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## Grants, Fellowships and Awards

### Grants

- 2023– DFG research grant AN 1545/4-1, project "Equivariant and weak orientations in the motivic homotopy theory", 226.650 euro. Joint with Nikita Geldhauser.
- 2022– DFG Heisenberg Program, grant AN 1545/1-1, project "Motivic homotopy theory and cohomological properties of algebraic varieties", 540.000 euro.
- 2021–2022 "Basis" Foundation grant "Stable motivic homotopy theory, K-theory and generalized motivic cohomology", Principal Investigator, 1.200.000 RUB/year.
- 2020–2022 joint RSF-DFG grant 20-41-04401 "Algebraic bordism spectra: Computations, filtrations, applications", Principal Investigator on the Russian side, 6.000.000 RUB/year on the Russian side. Principal Investigator on the German side is Oliver Röndigs.
- 2014–2015 RFBR grant 14-01-31095-mol\_a "*Homotopical methods in algebraic geometry and derived Witt groups*", Principal Investigator, 400.000 RUB/year.
- Investigator RFBR grants 19-01-00513-a, 18-31-20044-mol\_a\_ved, 16-01-00750-a, 15-01-03034-a, 13-01-00429-a, 12-01-33057-mol\_a\_ved, 12-01-31100-mol\_a, 10-01-00551\_a, RSF grant 14-11-00456  
(not PI)

### Fellowships and Awards

- 2020 Simons Foundation Fellowship to stay at Isaac Newton Institute, Cambridge
- 2018–2020 "Young Russian Mathematics" fellowship, 480.000 RUB/year
- 2017 Fellowship to stay at Mittag-Leffler Institute, Stockholm
- 2015 IAS fellowship to stay at Institute for Advanced Study, Princeton
- 2013–2015 Dynasty foundation fellowship for Russia's young mathematicians, 360.000 RUB/year
- 2013, 2015 Gazprom Neft prize for young mathematicians
- 2012 Young Mathematician prize of the St. Petersburg Mathematical Society
- 2010, 2012 Rokhlin prize for St. Petersburg young mathematicians

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## Mentorship and Research Direction

- 2026 Andriy Kryzhanovskyy, MSc student at LMU Munich
- 2025–2026 Anton Lin, MSc student at LMU Munich
- 2022– Egor Zolotarev, PhD student at LMU Munich
- 2021–2022 Tariq Syed, Postdoc at EIMI

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## Services to the community

- Refereeing Adv. Math., Algebr. Geom. Topol., Ann. Inst. Fourier, Ann. K-theory, Compos. Math., Doc. Math., IMRN, Invent. Math., Izv. Math., J. Algebra, J. Inst. Math. Jussieu, J. Pure Appl. Algebra, J. Topology, Math. Proc. Cambridge Philos. Soc., Math Z., Pacific J. Math., St. Petersburg Math. J., Transform. Groups, MathSciNet
- Reviewer German Research Foundation
- Organizing Zoom seminar Zoom seminar on  $\mathbb{A}^1$ -topology, motives and K-theory in 2020-2022  
<https://indico.eimi.ru/category/12/>.
- Organizing conferences **Motifs and Motives**, June 29 – July 03, 2026 at LMU, Munich, Germany. Organizers: Alexey Ananyevskiy, Nikita Geldhauser, Andrei Lavrenov and Maksim Zhykhovich. [Webpage](#)
- Emerging Research in Algebraic Groups, Motives, and K-Theory**, September 9-13, 2019 at EIMI, St. Petersburg, Russia. Organizers: Alexey Ananyevskiy, Anastasia Stavrova and Nikolai Vavilov. [Webpage](#)
- Petersburg motives (dedicated to Ivan Panin's 60th birthday)**, September 2-6, 2019 at EIMI, St. Petersburg, Russia. Organizers: Alexey Ananyevskiy, Alexander Merkurjev and Serge Yagunov. [Webpage](#)
- Motives in St. Petersburg 2018**, September 3-14, 2018 at EIMI, St. Petersburg, Russia. Organizers: Alexey Ananyevskiy, Grigory Garkusha and Ivan Panin. [Webpage](#)
- Moscow – St. Petersburg algebraic geometry symposium for young mathematicians**, May 10-11, 2018 at Chebyshev Laboratory, SPbU, St. Petersburg, Russia. Organizers: Alexey Ananyevskiy, Valery Gritsenko, Peter Zograf. [Webpage](#)